

COMMON QUESTIONS

| Question |     |       |    | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks available |     |     |       |       |      |
|----------|-----|-------|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |       |    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 8/1      | (a) | (i)   |    | naphtha                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |     | 1   | 1     |       |      |
|          |     | (ii)  |    | 20<br>accept C <sub>20</sub> / C <sub>20</sub> H <sub>24</sub>                                                                                                                                                                                                                                                                                                                                                                                                        |                 |     | 1   | 1     |       |      |
|          | (b) | (i)   |    | C <sub>5</sub> -C <sub>8</sub> accept petrol                                                                                                                                                                                                                                                                                                                                                                                                                          |                 |     | 1   | 1     | 1     |      |
|          |     | (ii)  |    | supply is greater than demand for all fractions <input type="checkbox"/><br>supply is greater than demand up to C <sub>16</sub> after which demand is greater than supply <input type="checkbox"/><br>demand is greater than supply up to C <sub>16</sub> after which supply is greater than demand <input checked="" type="checkbox"/><br>the difference between supply and demand increases up to C <sub>16</sub> after which it decreases <input type="checkbox"/> |                 |     | 1   | 1     |       |      |
|          |     | (iii) | I  | C <sub>8</sub> H <sub>18</sub>                                                                                                                                                                                                                                                                                                                                                                                                                                        |                 |     | 1   | 1     |       |      |
|          |     |       | II | (addition) polymerisation                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1               |     |     | 1     |       |      |

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks available |     |     |       |       |      |
|----------|-----|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          | (c) |  |  | award (1) for each correct method <b>and</b> factor removed<br><br>water – heat removed<br><br>'beating' – air/oxygen removed<br>accept foam – air/oxygen removed<br>accept dropping powder (from aeroplane) – air/oxygen removed<br>do not accept CO <sub>2</sub> / fire blanket / any fire extinguisher<br><br>fire breaks – fuel removed<br>accept bulldozing trees / cutting down trees / back burning<br><br>award (1) for three correct methods if no other mark awarded |                 | 3   |     | 3     |       |      |
|          |     |  |  | Question 8/1 total                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1               | 3   | 5   | 9     | 1     | 0    |